K.H.U. Faculty of Engineering and Natural Sciences

FENS 401-402 BACHELORS PROJECT
PROPOSAL

SLOPE-AWARE SHORTEST PATH MOBILE NAVIGATION APPLICATION FOR PEDESTRAINS

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Submission Date: 30.12.2025

ABSTRACT

Currently, widely used pedestrian navigation applications mostly generate routes based on the shortest distance or fastest time criteria, taking into account the effects of slope and elevation changes on human physiology to a certain extent. However, in topographically complex cities like Istanbul, a short route can require significant physical effort, while a longer route with a more balanced slope can be much more comfortable for the user. This project aims to develop a slope sensitive route planning approach that considers slope and elevation as a fundamental decision variable in pedestrian focused navigation systems. In this study, OpenStreetMap road network data will be integrated with Digital Elevation Models (DEMs) to create asymmetric and physiologically based cost functions for each road segment. Using models adopted from different studies, such as Tobler's Walking Function and Langmuir's version of Naismith's Rule, the effect of slope on walking effort will be expressed as a weighted graph. A*, Dijkstra, and Bidirectional Search algorithms will be applied to the resulting slope weighted graphical structure to generate alternative routes with the shortest, fastest, and most balanced slopes. The project aims to develop a cross platform mobile application that allows users to visualize route gradient profiles and difficulty levels.

AIM

In large cities like Istanbul, where urban planning is inadequate, one of the biggest problems is getting from one place to another, or even being able to walk. The goal of well-known navigation tools like Apple Maps, Yandex Maps, and Google Maps is to assist users in this way. In walking or cycling modes, these apps prioritize getting the user to their destination by the shortest path, or they provide alternate routes in addition to the shortest path. Although these apps take inclination information into account when calculating routes, they make very little use of it. They merely provide the user with the option to reduce the height differential along the route [1-3].

Studies have shown that uphill climbing requires higher physiological effort compared to walking on flat terrain. Conversely, it has also been proven that descending steep slopes can have negative effects such as muscle strain and loss of balance. Routes that require continuous uphill or downhill movement put strain on the musculoskeletal system by subjecting the body to a uniform load for a long period of time. These routes also cause people to tire more quickly. Routes with a balanced elevation profile along the way reduce physiological and metabolic load by spreading energy expenditure over time. Furthermore, these routes increase walking comfort and sustainability. In other words, routes where the slope increases and decreases more gradually and evenly improve the load distribution between different muscle groups. Routes with balanced slopes limit sudden increases in heart rate and allow for more efficient energy use [4–6].

Digital Elevation Models (DEMs) can be used to gather altitude information for several locations worldwide. These models are among the most important data sources used in navigation systems to improve terrain accuracy. Basic pathfinding algorithms like Dijkstra and A* are used by many navigation applications, but their use of uniform edge weights ignores the challenges posed by the slope of the path, making their basic versions inadequate. Therefore, these algorithms are being improved and used by current navigation applications to account for asymmetric costs. However, even these applications aim to reach the desired point either via the shortest path or with minimal incline. By reducing sudden changes in slope throughout the route, it is necessary to design routes that are more compatible with human physiology and metabolism [7–11] .

The goal of this project is to create and deploy an intelligent walking-based navigation system that minimizes abrupt changes in elevation along the trip and the amount of physical effort needed by pedestrians while determining the most effective path in terms of time and distance. In addition to viewing comprehensive route information, users will have access to many route options, including flat and sloping roads. Road network data will be obtained using OSMnx, and elevation values will be obtained using the Digital Elevation Model. The project will primarily focus on the European side of Istanbul. Each road segment will have an asymmetric effort-based cost computation that combines geometry and elevation. Pathfinding algorithms such as A*, Dijkstra, and Bidirectional Search will be developed to be sensitive to gradient and the gradient distribution along the route. This aims to minimize elevation changes encountered along the way. Users will be able to reach their destination via the fastest, shortest, or least gradient route [12-14].

Finally, a mobile application will be developed where users can see elevation changes along the route and the difficulty level of the route in terms of slope. This mobile application will have an intuitive interface and cross-platform support. Ultimately, it will provide a user-friendly navigation experience offering alternative routes to meet different user needs.

Research Questions, Hypothesis, and Validation Tests

The following is a summary of the research questions (RQs), Hypothesis' and, Validation Tests:

RQ1: How can pedestrian road networks and elevation raster data be precisely combined into a single weighted graph?

Hypothesis : GIS functions including CRS transformation, point sampling, and raster-vector mapping will be performed using GeoPandas and Rasterio to consistently connect each edge with thorough elevation data.

Validation Test: The integration pipeline will be validated by checking that the CRS is correctly aligned and that a significant number of road edges have valid elevation/slope properties after the raster sampling and mapping steps. The coverage ratio and a few spot checks of the raw DEM values at the same locations will be used for this purpose.

RQ2: How should the effect of uphill or downhill movement on the physical cost of walking be mathematically modeled?

Hypothesis: A slope-sensitive formula will be designed considering the slope percentages of the paths and the physiological effort they require. Based on current research, different weighting parameters will be simulated and tested to find the most suitable formula.

Validation Test: Various slope-sensitive cost expressions and values of parameters will be compared through analysis of routes obtained for a common set of origin-destination points. Route solutions will be evaluated using effort-related metrics obtained from the graph, such as elevation gain, average and maximum slope, and a measure of estimated effort, to determine which expression of consistent formulation is best.

RQ3: Which shortest path algorithms provide the best performance and result quality for slope-sensitive route calculation on large-scale maps?

Hypothesis: Improved Dijkstra, A*, and Bidirectional Search algorithms will be implemented in C++, and their routing results and execution times will be compared through testing. Based on the results, the algorithms will be modified and further developed.

Validation Test: A* Search, Dijkstra, and Bidirectional Search algorithms will be compared using repeated routing queries performed on the same map, based on execution time and optionally memory consumption, and their routing costs will be checked for consistency, using Dijkstra algorithm as a standard of optimality.

RQ4: In comparison to conventional shortest-time routes, how do slope-aware routes enhance user comfort and decision-making?

Hypothesis: Differences in elevation gain, distance, segment slope, and estimated energy consumption will be compared, and tests will be conducted with real users.

Validation Test: The slope-sensitive routes shall then be compared with the traditional shortest-distance routes using quantifiable elements that shall already be set for evaluation: route distance, cumulative elevation gain, average slope index, maximum slope index, and effort score value. If possible, a limited evaluation shall then be done using user-based testing for basic comfort/preference data as additional evidence for validation purposes.

RQ5: How should elevation and difficulty information be displayed in a mobile navigation interface?

Hypothesis : Slope heatmaps and elevation profiles are examples of visual indications that will be used in the design of the user interface. The interface will be improved in response to user feedback.

Validation Test: The mobile UI user experience will be assessed for validation through functional testing and usage testing, with user feedback solicited for general usability metrics related to navigating and using routes. Comparison of views, such as slope displays and summary displays of elevation and efforts, will be metrics-tested through user assessment of their ability to determine routes based on difficulty displays.

METHODOLOGY:

This section describes the overall methodological approach used in the project. It describes the data sources, preprocessing steps, modeling approaches, and algorithmic techniques that will be used to construct a slope-aware pedestrian navigation system.

1. Data Acquisition and Preprocessing

The data acquisition and preprocessing steps are required to construct a reliable slope-aware pedestrian network. In this part, pedestrian road networks and elevation data will be collected, prepared and integrated into each other to ensure consistency and suitability for slope-based analysis and routing.

1.1 Road Network Extraction

OSMnx is a Python package used to extract pedestrian-accessible street networks from OpenStreetMap and convert them into NetworkX graphs. As shown in Figure 1, the generated geometry is converted into a directed multigraph while preserving intersection nodes and walkable sections. Topological cleaning eliminates inaccessible branches, self-loops, and redundant edges. Sırımcılar Street and Gemiciler Street, shown in Figure 2, are examples of these self-loops [12-13].

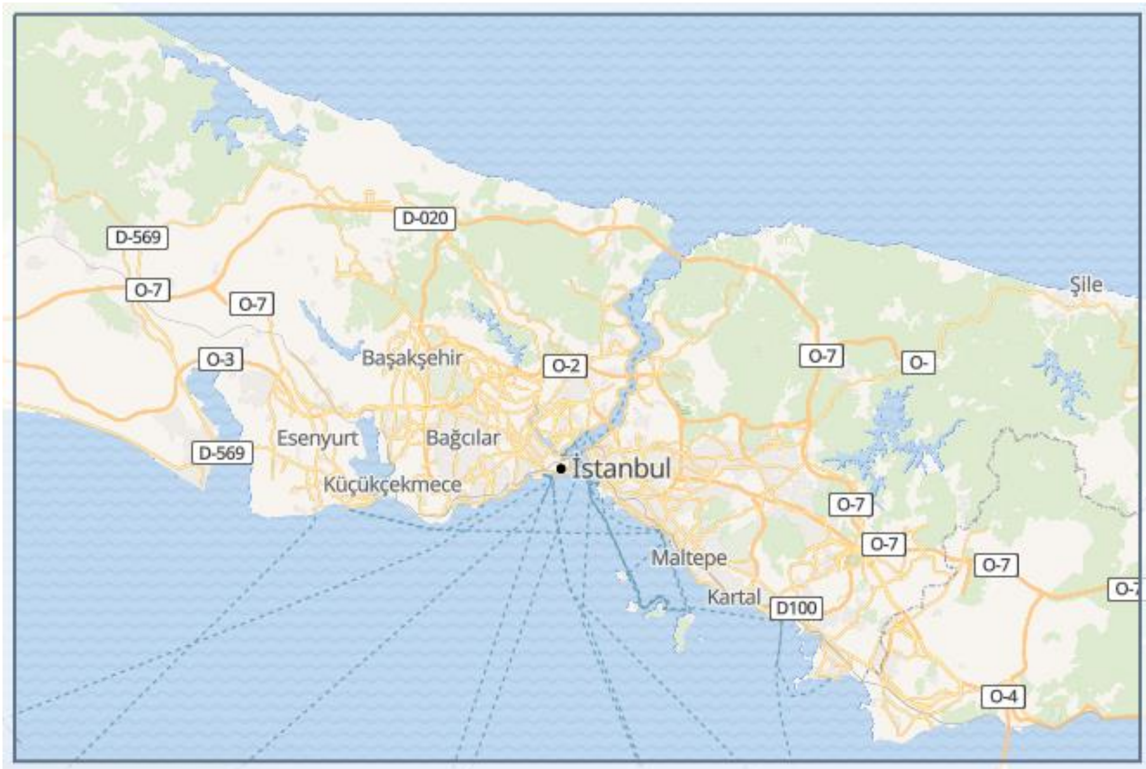


Figure 1. Road map of the Istanbul [14]

Figure 2. Self-Loops Road Example [1]

1.2 Digital Elevation Model (DEM)

In this project, elevation data was obtained using a high-resolution Digital Elevation Model (DEM). DEMs are raster-based data structures that divide the terrain into grid cells, allowing access to the elevation information of each cell. They are used for slope, elevation difference, and topographic cost calculations. [14]

Since OSMnx data is a discrete and topological vector graphic, and the raw DEM data is a continuous raster structure, the DEM data requires a series of preprocessing steps before working with these two datasets [15-16]. The data formats must be interoperable; therefore, the DEM data must be converted to a metric Coordinate Reference System (CRS) to accurately calculate distances. This conversion is critically important for creating slope and distance-based cost functions. As shown in Figure 3, to be compatible with the project, this data is trimmed to be limited to the European side of Istanbul. This ensures that only the areas necessary for the analysis are processed, reducing calculation costs.

Figure 3. Digital Elevation Model (DEM) of the European Side of Istanbul [14]

Preprocessing the DEM data yields a GeoTIFF raster file showing the elevation value of each grid cell. GeoTIFF is an extended version of the Tagged Image File Format (TIFF) that includes georeferencing. A GeoTIFF file is data consisting of pixels that carry the coordinate reference system and projection information of the image. This allows the raster data to be matched with its actual location on the map. This data can then be processed using the Rasterio library in the Python environment. Thanks to the Rasterio library, height sampling can be performed for node and edge geometry data, i.e., road segments, on the road network (OSMnx) [17].

The DEM is prepared for sampling along road geometry in the future. Each section of the road should be separated into sample locations that are regularly spaced (e.g., 5–10 m intervals) and the interval that will yield the best results should be identified. Each point's elevation should obtained from the DEM using bilinear interpolation. Each edge thus has a constant elevation profile. The formula for calculating slope is as follows:

$$s = \frac{\Delta h}{d} \quad (1)$$

where s is slope, h is elevation, and d is segment length.

Both positive (uphill) and negative (downhill) slopes are maintained to account for asymmetric traversal costs.

3. Effort-Based Asymmetric Cost Modeling

Modern commercial navigation systems like Google Maps, Apple Maps, and Yandex Maps primarily use distance and time-oriented cost functions on two-dimensional road networks for route calculations. However, there is no clear documentation regarding the structure of these cost functions and the algorithms used. These systems model the road network as a topological graph, calculating the shortest or fastest routes based on nodes and edges. At least as far as our research has shown, although these platforms possess high-resolution digital map infrastructures and global topographic datasets, there is no publicly available, detailed methodology for using slope information as a primary decision variable in route optimization. Slope data is mostly evaluated in the context of estimated time, speed profiles, or visual information (Figure 4-7) presented to the user [18-22].

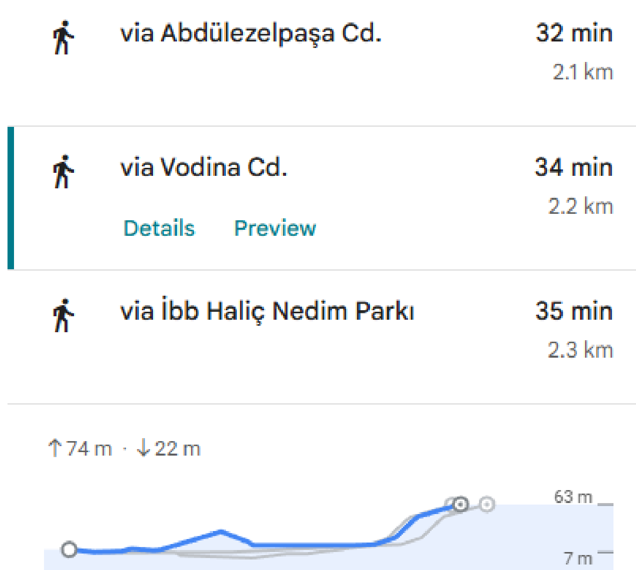


Figure 4. Google Maps' Route Suggestions [1]

Google Maps enhances slope awareness by providing elevation profiles to the user, especially in pedestrian and cyclist navigation; Apple Maps similarly provides visualized elevation and slope information during the route preview phase. However, neither system has explicit technical documentation indicating that the routing algorithm directly uses a cost function that minimizes slope or optimizes energy consumption. Yandex Maps stands out, particularly in urban areas, for its detailed road and traffic data, but explanations regarding how gradient information affects route selection are mostly given through indirect performance indicators (time estimation accuracy, traffic adaptation). This suggests that gradient is often treated as a secondary or implicit variable [23-26].

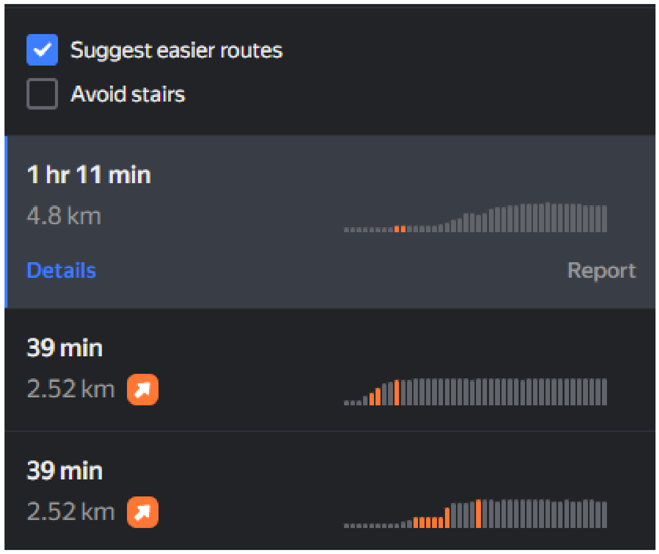


Figure 5. Yandex Maps' Route Suggestions [2]

Unlike commercial navigation systems, the approach followed in this study treats gradient not merely as a visualization or time estimation improvement tool, but as a fundamental physical parameter directly determining route cost. Due to the closed-source and commercial nature of systems like Google, Apple, and Yandex, their detailed algorithmic decision-making processes are not transparent; therefore, they are generally considered "gradient-aware" systems in the literature. Targeting this gap, the proposed method aims to go beyond the classical distance and time-based optimization understanding offered by existing commercial systems by presenting a model that explicitly calculates gradient and integrates it into the route selection process with asymmetric and nonlinear cost functions.

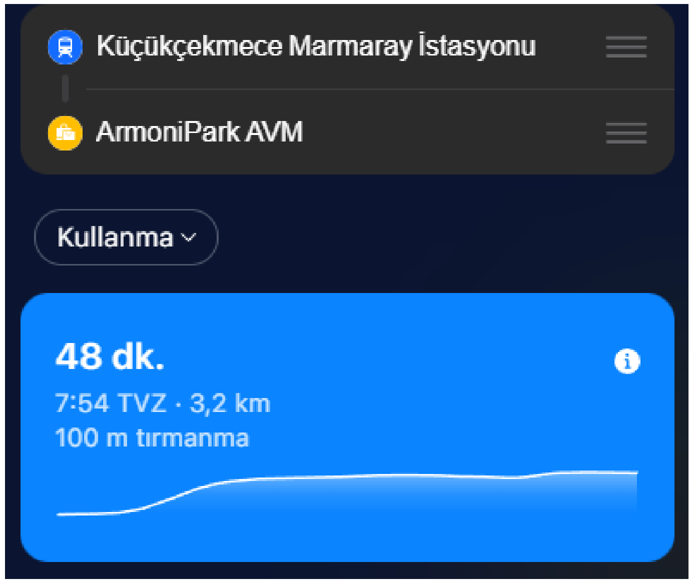


Figure 6. Apple Maps' Route Suggestion that Uses Hills Option [3]

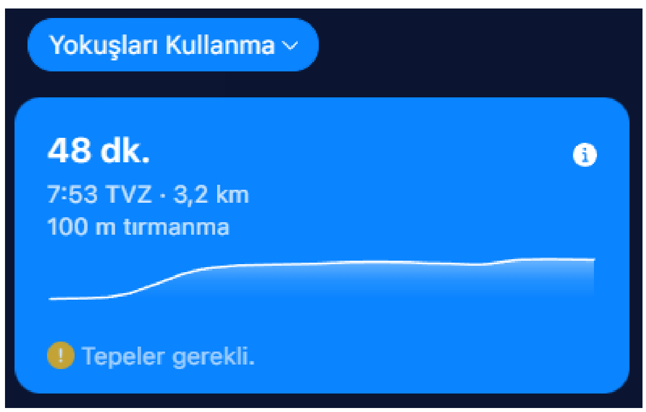


Figure 7. Apple Maps' Route Suggestion that Avoid Hills Option [3]

3.1 Tobler’s Hiking Function

In this project, we primarily consider two methods for integrating the gradient of the trail network into the route cost function. These are Tobler’s Hiking Function, which models the change in speed and energy of human movement depending on topography, and Naismith’s Rule, developed by Langmuir. Tobler’s Hiking Function defines walking speed as an exponential function of the gradient:

$$v(s) = 6 \times e^{-3{,}5 \times |S + 0{,}05|} \quad (2)$$

where v(s) is walking speed (km/h) and s is slope (rise/run).

Walking effort (cost) for a segment is defined as time travel:

$$C = \frac{d}{v(s)} \quad (3)$$

where C is cost and d is segment length.

It states that speed increases especially on gentle descents and decreases on steep ascents and descents. Because this model treats the gradient as a continuous variable, it allows the local gradient values calculated via DEM to be directly converted into an edge-based cost coefficient. Thus, the trail length is scaled with the gradient-dependent speed reduction, yielding a more realistic “effective distance” or time cost [27].

3.2 Naismith's Rule

Naismith's Rule, on the other hand, is a more intuitive model initially proposed for mountain hikes and has been expanded with Langmuir's contributions to include the effect of descent gradients. According to this approach, in addition to the horizontal distance, a fixed time penalty is defined for each specific amount of ascent; It is assumed that speed will decrease again on steep descents. The Naismith–Langmuir model is computationally simpler and is suitable for adding the slope as a discrete penalty term to the cost function. In this study, these two methods are evaluated as alternative cost models representing the physical effect of the slope on the route at different levels of detail; the Tobler approach offers a continuous and smooth cost surface, while the Naismith–Langmuir approach provides a more interpretable and intuitive cost definition. [30]

The basic Naismith Rule recommends adding 1 hour for every 5 km on flat terrain, plus an additional 1 hour for every 600 m climb. While the basic Naismith rule completely ignores descents, Langmuir demonstrated that descent gradient has a nonlinear effect.

Langmuir acknowledges that descents with gradients of approximately 5–12% accelerate walking. In such controlled descents, gravity supports movement, the stride rhythm is not disrupted, and energy expenditure is reduced. Therefore, it is recommended to deduct a certain amount of time from the time predicted by Naismith in this gradient range. In other words, the descent is considered a negative cost in the cost function.

Langmuir also stated that the advantage of descending disappears when the slope exceeds 12-15%. The risk of loss of balance, the need for braking, and musculoskeletal load increase. Langmuir states that in this case, walking speed decreases and the time should be increased again. Therefore, steep descents, just like uphill climbs, generate a positive cost.

Langmuir's most important contribution is his explicit definition of the non-symmetric slope. An ascent and descent with the same absolute slope are not equivalent in terms of time and effort. This approach is an early precursor to the asymmetric cost function concept used in modern route optimization and forms the theoretical basis for assigning different weights to ascent and descent edges in graph-based algorithms [28].

In summary, Langmuir improvements offer a piecewise model that generates bonuses or penalties depending on the slope range, rather than viewing the descent as a uniform "convenience". Therefore, it is more discrete and rule-based compared to the Tobler function; however, it is still preferred in navigation and GIS applications due to its low computational cost.

4. Routing Algorithms

Routing algorithms are used to determine the optimal path between nodes defined on a graph and the edges connecting those nodes. While this optimal path varies depending on the problem, in this project it can be defined as the shortest distance, the lowest time, the lowest cost, or the lowest topographic cost, including slope and elevation.

4.1 Graph Structure

The directed graph that represents the slope-weighted pedestrian network is appropriate for shortest-path methods. Previous research demonstrating that sophisticated methods (e.g. Contraction Hierarchies) can significantly reduce search space and query time in big road networks serves as motivation for efficient routing on such graphs [29].

4.2 Comparative Analysis of Dijkstra, A* search, and Bidirectional Search Algorithms

Route calculation is one of the key components of this project, given that the cost of routing is not just a function of distance but also involves elevation and effort. The pedestrian graph is represented as a weighted graph that is directed, with each edge denoting effort that is slope-sensitive instead of the more common length-sensitive approach. Such a scenario influences route calculation algorithms significantly, especially with respect to their efficiency on mobile devices.

In this section, a brief comparison of the suitability of Dijkstra, A* search, and Bidirectional search algorithms to slope-sensitive pedestrian routing is given.

Dijkstra's Algorithm

It calculates the minimum path cost by exhaustively traversing the graph without taking into consideration the direction of the destination. This algorithm is guaranteed to provide the optimum solution, even with complex cost or effort functions that exhibit asymmetry through slope considerations.

However, since it deals with a lot of nodes, it becomes computationally intensive. This is especially true for densely populated areas. Hence, the application of Dijkstra's algorithm is limited in this project only for purposes of comparison. It is not the primary means for navigation [30-32].

A* (A-Star) Search

A* is an improvement over Dijkstra's algorithm since it allows the search path to converge on the destination using a heuristic function. In the current project, the heuristic is determined by the straight-line distance, presuming a flat terrain, which is a safe low-bound on pedestrian efforts.

This helps in significantly reducing the space of search while still ensuring that it is able to obtain optimized solutions. It is for this reason that A* is chosen as the major routing algorithm for the application since it is more suited for real-time pedestrian routing applications for mobile devices [34-36].

An enhanced version of the A* algorithm also exists, which heuristically incorporates the number of nodes. In this modified algorithm, Dijkstra's optimum and heuristic search efficiency are combined in A*. This is consistent with research highlighting A* as a high-performance pathfinding technique when heuristic methods are well-designed. The heuristic method incorporates Euclidean distance scaled according to expected height-based speed to reduce the extended number of nodes [10].

Bidirectional Search

Bidirectional search is then used to further refine the routing paths by concurrently conducting two searches: one from the initial position and another from the destination position. Once both search boundaries converge, the path is formed.

This approach significantly cuts down on the number of nodes to be explored, thereby improving response time for both large and complex city graphs. Despite adding more implementation complexity, Bidirectional A* is recognized as an optimized sophisticated step in improving the efficiency that cannot be achieved by limited mobile hardware devices [33].

Table 1. Comparison of Dijkstra, A* and Bidirectional A* algorithms

Algorithm	Search Strategy	Optimality	Computational Cost	Mobile Suitability	Role in This Project
Dijkstra	Uninformed, exhaustive search	Guaranteed	High (large search space)	Low	Used as a baseline for validation and comparison
A*	Heuristic-guided search	Guaranteed (admissible heuristic)	Moderate	High	Main routing algorithm for real-time navigation
Bidirectional A*	Dual-direction heuristic search	Guaranteed	Low–Moderate	Very High	Performance optimization for dense urban graphs

5. Minimizing Slope Fluctuations Along the Route

To reduce slope fluctuations, it is possible to consider not only the total distance or average slope values but also the continuity of the slope along the route as an optimization criterion in the route optimization problem. In this context, it can be suggested to quantitatively define the slope differences between successive route segments and include these differences as an additional penalty term in the cost function. Such an approach allows routes with short distances but high-frequency ascents and descents to be considered more costly compared to longer alternatives with slope changes spread over time and a more stable topographic profile. Through this method, the route generation process can be transformed into a multi-criteria optimization problem that not only minimizes the total amount of climb but also encourages gradual slope transitions.

6. Planned Application Features and Reference User Interface Designs

Elevation gain, gradient severity, and physical effort are examples of new sorts of information that are not included in conventional navigation applications and are introduc ed by a slope-aware navigation system.

In order to transmit complicated facts in an intelligible manner without overpower ing the user, the user interface (UI) must be properly designed.

The fundamental UI concepts and functional elements intended for the mobile application are described in this section [34-35].

6.1 Design Criterias

The UI design is mainly guided the criterias listed below:

- **Cognitive Simplicity:**
  Instead of presenting complex data as numbers, use visual cues.
- **Progressive Disclosure:**
  Present the most important details initially, with further in-depth analysis accessible upon request.
- **Accessibility-First Design:**
  Make sure that users who are colorblind, old, or have limited mobility can utilize the product.
- **Real-Time Responsiveness:**
  UI components must refresh seamlessly without interfering with the flow of navig ation.

6.2 Core Application Features

The application we aim to build will have several core concepts. It has to visualize the road the user currently walk by taking the slope into account. Also the summarization of the road should be given with calori calculations and avarage time. Alongside theese features, users should be able to share their personal routes in the community

6.2.1 Multi-Profile Route Selection

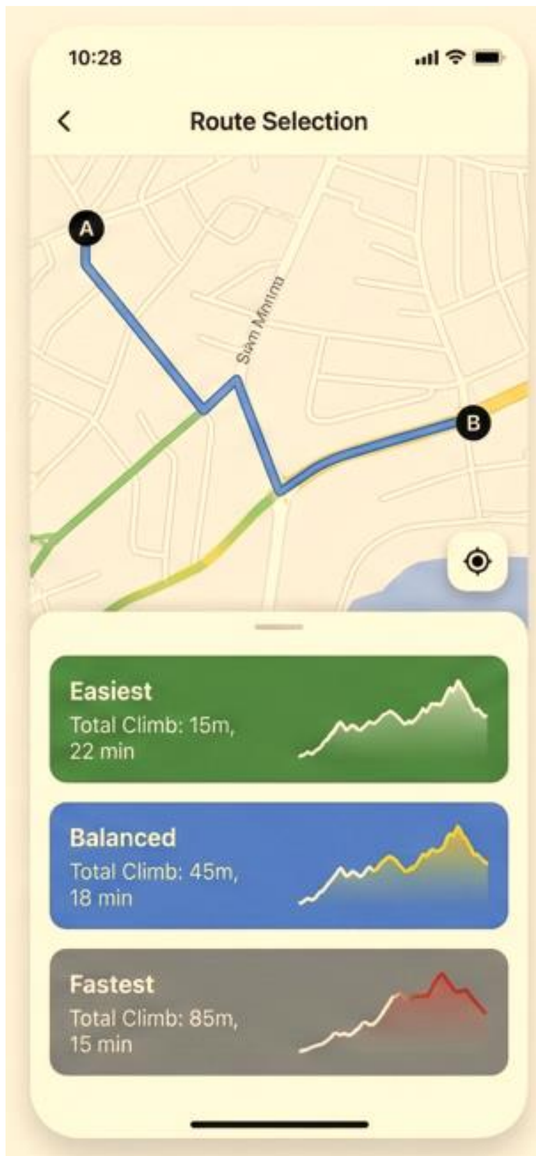


Figure 8. Routing selection screen with three options

Users can select different navigation profiles based on their physical condition or preference as illustrated in Figure 8:

- **Easiest Route:** Minimizes physical effort and slope.
- **Balanced Route:** Optimizes both distance and elevation.
- **Fastest Route:** Prioritizes shortest travel time.

Each route option is displayed as a **card-based layout** showing:

- Total distance
- Estimated walking time
- Total elevation gain/loss
- Effort score (visualized using icons rather than raw energy units)

6.2.2 Slope-Aware Route Visualization

Figure 9. Route visualization according to the slope rates

This approach employs gradient-coded polylines in contrast to conventional navigation apps that show routes using a single color:

- Green: Low slope (comfortable walking)
- Yellow: Moderate incline
- Red: Steep ascent
- Dark or dashed segments: Stairs or extreme slopes

This allows users to instantly identify physically demanding sections before starting the route which is shown in Figure 9.

6.2.3 Interactive Elevation Profile

The application includes a dynamic elevation graph that shows:

- Horizontal axis: Distance along the route
- Vertical axis: Elevation

Interactive Feature:

When the user drags a finger along the elevation profile, the corresponding position is highlighted on the map. This enables users to locate steep climbs or descents precisely.

6.2.4 Accessibility and Safety Features

To support vulnerable users, the app includes:

- For those with visual impairments, there is a high contrast mode.
- In addition to color labeling, there are visual pattern indicators.
- Automatic warnings for slopes exceeding predefined safety thresholds.
- Optional avoidance of stairs and extreme gradients.

Figure 10. Real-time map updates with paths slope rates

6.3 Navigation and Feedback During Walking

During active navigation:

- Real-time location tracking is displayed.
- Upcoming slope changes are announced visually and optionally via vibration alerts.
- Users receive advance warnings before steep inclines or descents.

6.4 Overall User Experience Objective

Apart from the core navigation functionality, the application also features a social aspect that is aimed at promoting user engagement with the application. This technology allows users to make their travels more engaging by sharing moments and experiences that develop from their journeys, apart from determining the routes.

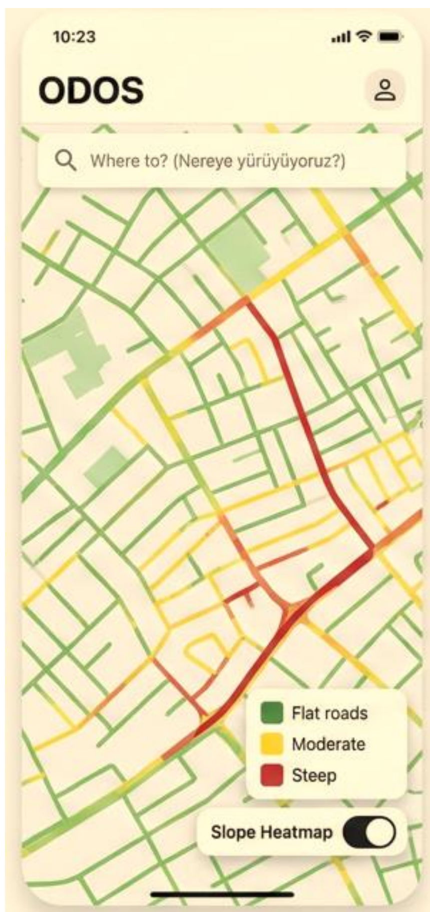


Figure 11. Overall heat map slope-aware visualization

6.5 Social Interaction and Community Sharing Module

In addition to its basic navigation features, there is a social aspect that allows for user interactions with the end aim of promoting frequent use of the application. This technology creates an interesting way of navigating, as it allows users to share moments of their journey, as opposed to just determining routes, shown in Figure 12

Figure 12. User-shared slope-aware walking paths screen

From a user interface perspective, social content is designed to remain optional and secondary. Photos and posts are displayed as small, map-based indicators that do not interfere with active navigation. Furthermore, a separate "Community" section enables browsing shared content by location, recency, or popularity without cluttering the main routing screen. Users remain in control of visibility through basic privacy settings that let them keep content private or share it as they see fit (Figure 13).

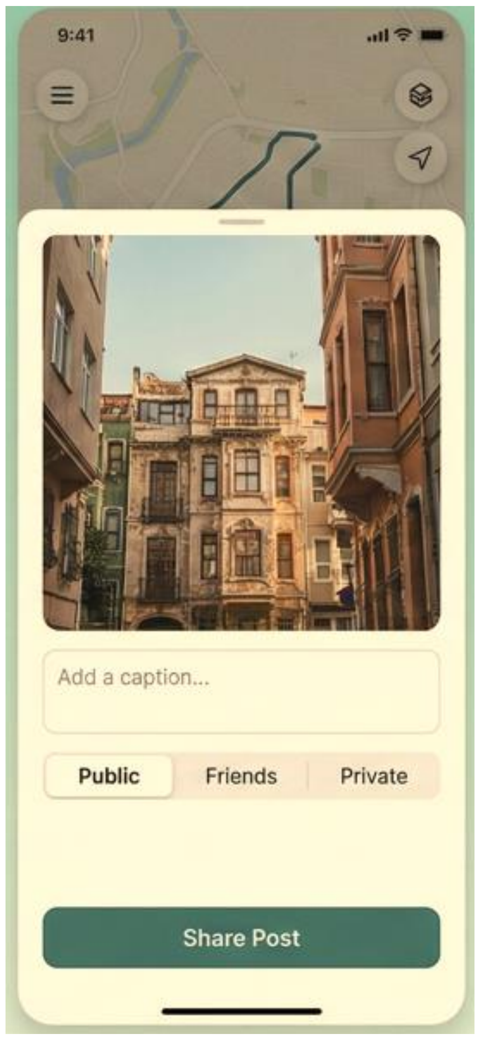


Figure 13. Walking path post creation screen

6.6 Calorie Expenditure and Physical Effort Feedback Interface

The software provides a social component in addition to its core navigation features, which are intended to make sure user interaction and continuing use. The software turns ordinary navigation into a more captivating and community-friendly activity, by allowing users to share events and experiences from their journeys alongside with calculating routes.

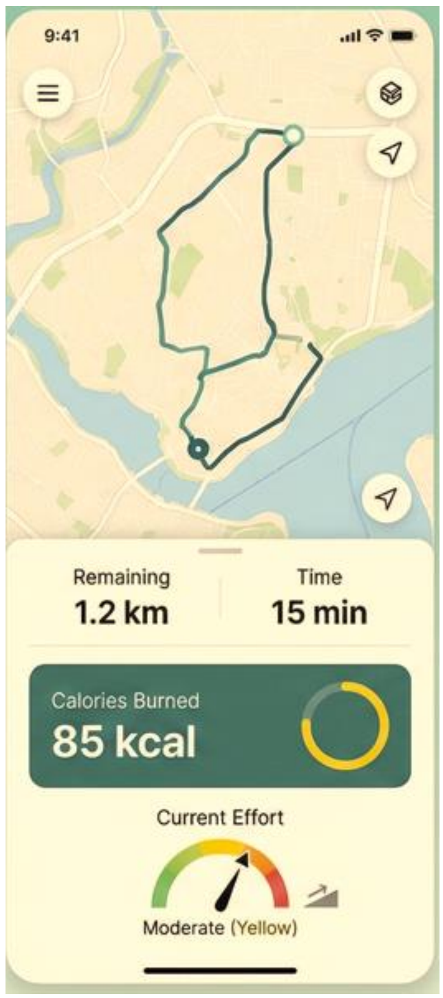


Figure 14. Slope-aware real-time route tracking interface.

Users are presented with an estimated calorie burn, distance, elevation gain, and total effort level prior to beginning a journey which can be seen in Figure 15. Users can observe how physical activity increases throughout their walk by gradually updating their calorie consumption throughout navigation. The average effort intensity, the total number of calories burned, and a comparison with walking the same distance on flat ground are displayed on a summary screen once the voyage is complete (Figure 14).

Figure 15. Post-walk summary with elevation and calorie metrics.

We plan to use Flutter or Expo react native for entire frontend development as they pros and cons in various aspects. While both technologies enable us to develop cross-platform, the decision should be taken in debugging as we don't have a chance to iOS debug for Flutter. However, with expo react native we could debug with our current devices during the cross-platform development [38].

Flutter and Expo React Native are both popular frameworks for building cross-platform mobile applications, but they follow different approaches. Flutter uses the Dart programming language and provides its own rendering engine, which allows developers to create highly consistent and smooth user interfaces across platforms. Expo React Native, on the other hand, is based on JavaScript and React, making it easier to learn for developers with web development experience. Expo simplifies the development process by offering many built-in APIs and tools, which speed up prototyping and deployment. While Flutter may offer slightly better performance in complex UI scenarios, Expo React Native stands out for its fast development cycle, large ecosystem, and strong community support [36-37].

WORK PACKAGE DESCRIPTION

Number and name of the package: Work Package I – Literature Review on Slope-Aware Pedestrian Navigation

Start date: 2025/11

Finish date: 2025/12

Short description: This work package focuses on conducting a literature review on pedestrian navigation systems, slope-aware routing, and effort-based pathfinding algorithms. The main goal is to create a theoretical foundation for our slope-sensitive navigation application.

Methodology: Reviewing relevant academic studies, technical reports, and popular navigation systems to understand ongoing approaches to pedestrian routing and slope-sensitive navigation.

In addition, existing navigation applications like Google Maps, Apple Maps, and Yandex Maps will be analysed to understand their approach on the topic of slope-sensitive navigation. Limitations of distance and time-based approaches will be reviewed to promote the proposed methodology.

Design of the cost functions, routing algorithms, and user interface implementations on the subsequent work packages will be guided by the findings of the literature review.

Outputs:

- Structured literature review summary
- Identification of gaps in existing pedestrian navigation systems
- Theoretical basis for slope-aware cost modeling and routing design

Relations to other packages: This work package provides the theoretical background for Work Package II (routing algorithms), Work Package III (backend computations), and Work Package IV (mobile application design), ensuring that all technical decisions are grounded in existing research.

Working team members (with contributions as percentages of the work packet):
Berkay Mustafa Arıkan (25%)
Mehmet Enes Oy (25%)
Nurettin Enes Karakulak (25%)
Fatih Tekçe (25%)

Work to be done by any body outside the team:
None

Number and name of the package: Work Package II – Construction of Slope-Aware Pedestrian Base Map

Start date: 2025/12

Finish date: 2026/01

Short description: This work package aims to generate a slope-aware pedestrian base map for Istanbul by combining pedestrian road network data with elevation data obtained from OpenTopography. The resulting map will form the core spatial dataset used throughout the project.

Methodology: Pedestrian road network data for Istanbul will be extracted from OpenStreetMap and filtered to include only walkable paths. Elevation data will be obtained from the OpenTopography portal in raster (GeoTIFF) format.

Using Python-based geospatial libraries such as GeoPandas, Rasterio, NumPy, and Shapely, the pedestrian road segments will be processed and interpolated at fixed spatial intervals. Elevation values will be sampled from the OpenTopography DEM for each interpolated point. Based on consecutive elevation samples, slope values will be computed for each road segment, capturing both average and maximum slope characteristics.

The processed pedestrian network will be enriched with distance, elevation gain, average slope, and maximum slope attributes. All information will be stored in a unified geospatial format suitable for routing algorithms and mobile application integration.

Outputs:

- Istanbul pedestrian road network enriched with elevation and slope attributes
- Slope-aware base map dataset derived from OpenTopography DEM
- Graph-ready geospatial data for routing operations

Relations to other packages: This work package provides the core spatial dataset required by Work Package II (routing algorithms), Work Package III (backend computations), Work Package IV (mobile application integration), and serves as a baseline dataset for evaluation in Work Package V.

Working team members (with contributions as percentages of the work packet):
Berkay Mustafa Arıkan (70%)
Mehmet Enes Oy (10%)
Nurettin Enes Karakulak (10%)
Fatih Tekçe (10%)

Work to be done by any body outside the team:
None

Number and name of the package: Work Package III – Advanced Slope-Aware Routing and Pathfinding Algorithms

Start date: 2026/01

Finish date: 2026/03

Short description: This work package focuses on developing slope-aware routing algorithms that utilize the elevation-enriched pedestrian map generated in Work Package I.

Methodology: Using the slope-aware map, asymmetric cost functions will be designed to model physical walking effort. Different penalties will be applied for uphill and downhill movements based on slope percentage and elevation gain.

The pedestrian network will be represented as a weighted graph, where edge weights reflect different optimization criteria such as shortest distance, minimum physical effort, and balanced routes. Classical shortest path algorithms such as Dijkstra and A* will be implemented and adapted to operate on the slope-aware graph using Python or C++ for performance optimization.

Outputs:

- Slope-based asymmetric cost functions
- Pedestrian routing algorithms supporting multiple route profiles
- Route comparison results (shortest vs easiest vs balanced)

Relations to other packages: Uses the slope-aware pedestrian base map generated in Work Package I and produces routing outputs consumed by Work Package III (backend analysis), Work Package IV (mobile application), and Work Package V (evaluation and validation)

Working team members (with contributions as percentages of the work packet):

Berkay Mustafa Arıkan (35%)

Mehmet Enes Oy (35%)

Nurettin Enes Karakulak (29%)

Fatih Tekçe (1%)

Work to be done by any body outside the team:

None

Number and name of the package: Work Package IV – Backend Algorithms for User Activity Analysis and Ranking

Start date: 2026/02

Finish date: 2026/04

Short description: This work package focuses on the design and implementation of backend algorithms that analyze user activity data and generate statistics and rankings within the application.

Methodology: User activity data such as selected routes, walked distance, elevation gain, slope difficulty, and walking duration will be collected during application usage. A structured backend data model will be designed to store user histories efficiently.
Statistical algorithms will compute aggregated metrics such as daily, weekly, and monthly walking summaries. Effort-based scores will be calculated using slope and elevation information obtained from routing algorithms. Based on these scores, leaderboard algorithms will rank users according to different criteria.
Data consistency, efficiency, and privacy considerations will be addressed throughout the backend design.

Outputs:

- Backend data schema for user activity records
- User statistics and effort score computation algorithms
- Leaderboard ranking and update mechanisms

Relations to other packages: Consumes routing and slope-related data from Work Package II and provides processed user statistics and rankings to Work Package IV. Outputs are evaluated and validated in Work Package V.

Working team members (with contributions as percentages of the work packet):

Berkay Mustafa Arıkan (35%)

Mehmet Enes Oy (35%)

Nurettin Enes Karakulak (29%)

Fatih Tekçe (1%)

Work to be done by any body outside the team:

None

Number and name of the package: Work Package V – Mobile Application Frontend Development and System Integration

Start date: 2026/02

Finish date: 2026/05

Short description: This work package focuses on developing the mobile application frontend and integrating routing and backend components into a unified user-facing system. The specific frontend technology (e.g., Flutter, Expo React Native, or similar frameworks) will be selected during the development process based on project requirements.

Methodology: A cross-platform mobile application frontend will be developed using a modern mobile application framework such as Flutter, Expo (React Native), or an equivalent technology. The application will allow users to select start and destination points, choose routing preferences, and request slope-aware routes generated by the routing module.

The frontend will visualize pedestrian routes on an interactive map using color-coded slope representations. Elevation profiles, route statistics, and comparative metrics will be presented to support informed route selection. User-related features such as activity summaries and leaderboards, generated by backend algorithms, will be integrated into the application interface.

The frontend will communicate with routing and backend components through well-defined APIs. Functional testing will be conducted to ensure correct data flow, usability, and system stability.

Outputs:

- Functional cross-platform mobile application frontend prototype
- Slope-aware route visualization and comparison screens
- User statistics and leaderboard user interfaces

Relations to other packages: Integrates outputs from Work Packages I (base map), II (routing algorithms), and III (backend algorithms) into a unified mobile application and provides user-facing results that are evaluated in Work Package V.

Working team members (with contributions as percentages of the work packet):

Berkay Mustafa Arıkan (9%)

Mehmet Enes Oy (40%)

Nurettin Enes Karakulak (50%)

Fatih Tekçe (1%)

Work to be done by any body outside the team:

None

Number and name of the package: Work Package VI – Evaluation, Validation, and Documentation

Start date: 2026/04

Finish date: 2026/05

Short description: This work package focuses on evaluating the correctness, performance, and usability of the developed system and preparing the final project documentation.

Methodology: The slope-aware pedestrian routing results will be evaluated by comparing them with traditional shortest-distance routes generated without elevation awareness. Quantitative metrics such as route length, total elevation gain, average slope, maximum slope, and estimated effort score will be analyzed.
Backend components such as user statistics and leaderboard algorithms will be validated using synthetic and real usage data to ensure correctness and consistency of computed rankings.
The mobile application will be evaluated through functional testing and scenario-based usage tests. Basic usability feedback will be collected to assess clarity of route presentation and user interaction.
All system components, algorithms, architectural decisions, and evaluation results will be documented in the final project report. Visual diagrams, flowcharts, and tables will be prepared to clearly present the system design.

Outputs:

- Quantitative evaluation and comparison results
- Validation reports for routing and backend algorithms
- Final project report and presentation materials

Relations to other packages: This work package evaluates and documents the outputs of Work Packages I, II, III, and IV.

Working team members (with contributions as percentages of the work packet):

Berkay Mustafa Arıkan (33%)

Mehmet Enes Oy (33%)

Nurettin Enes Karakulak (33%)

Fatih Tekçe (1%)

Work to be done by any body outside the team:

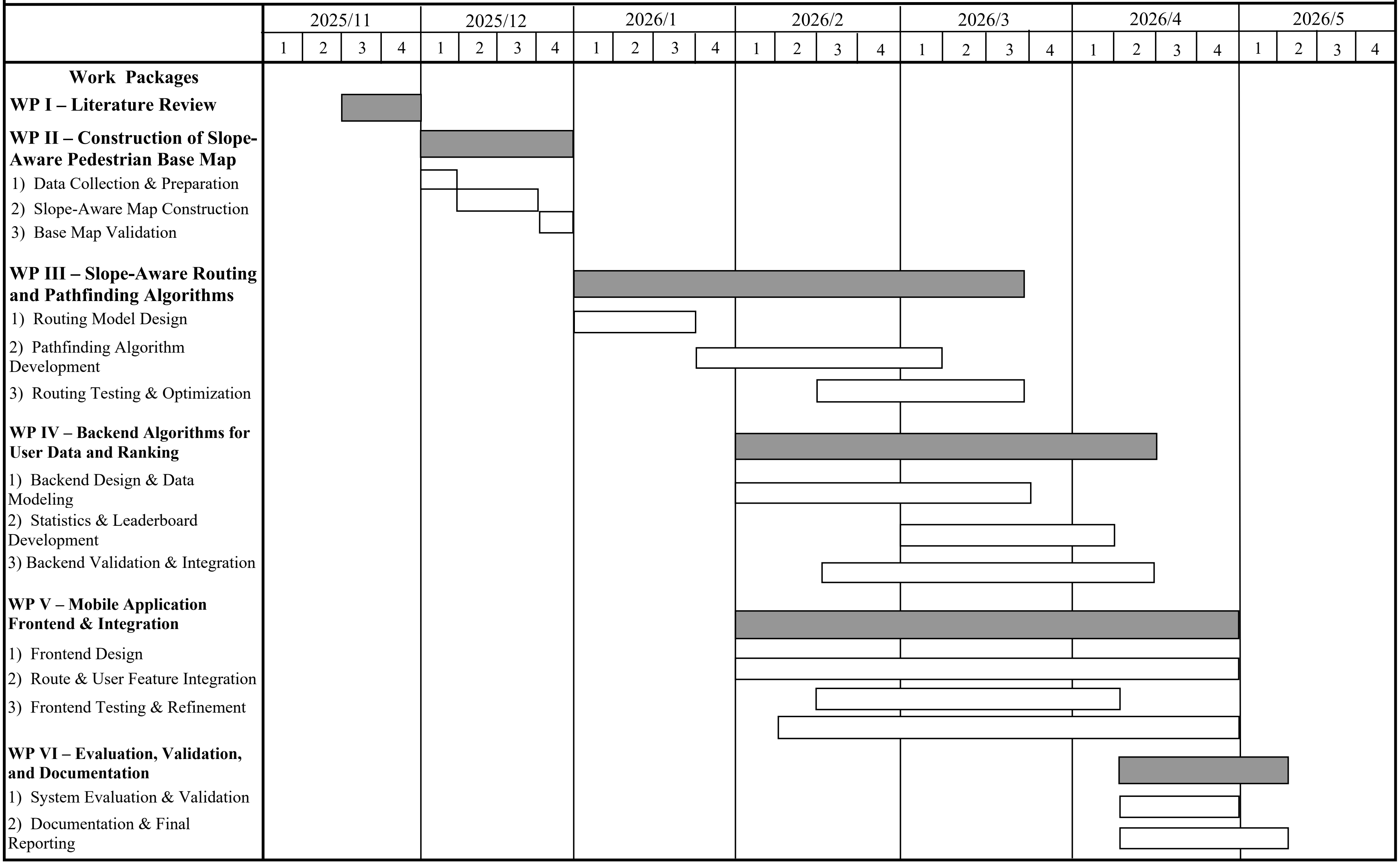
None

CONTRIBUTIONS OF THE TEAM MEMBERS:

Name, Surname	Work Packages												Total	
	I		II		III		IV		V		VI			
	h	%	h	%	h	%	h	%	h	%	h	%	h	%
Berkay Mustafa Arıkan	38	3,62	119	11,33	70	6,67	63	6,00	24	2,29	33	3,14	**309**	**33,05**
Fatih Tekçe	38	3,62	17	1,62	1	0,10	1	0,10	1	0,10	1	0,10	**21**	**5,62**
Mehmet Enes Oy	37	3,52	17	1,62	70	6,67	63	6,00	100	9,52	33	3,14	**283**	**30,48**
Nurettin Enes Karakulak	37	3,52	17	1,62	59	5,62	53	5,05	125	11,90	33	3,14	**287**	**30,86**
Total	**150**	**14,29**	**170**	**16,19**	**200**	**19,05**	**180**	**17,14**	**250**	**23,81**	**100**	**9,52**	**900**	**100**

PROJECT CALENDAR

PROJECT TITLE: **SLOPE-AWARE SHORTEST PATH MOBILE NAVIGATION APPLICATION FOR PEDESTRAINS**



INFRA-STRUCTURE REQUIREMENTS:

Laboratory(ies):

Equipment:

Computer(s):

Personnel:

PROJECT BUDGET:

	Work Packages				
	I	**II**	**III**	**IV**	**TOTAL**
Equipment	0	0	0	0	0
Consumable goods	0	0	0	0	0
Publications/Software	0	0	0	0	0
Transportation	0	0	0	0	0
Services	0	0	0	0	0
TOTAL	0	0	0	0	0

Equipment:

Consumable goods:

Publications/Software:

Transportation:

Services:

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